



The Empire Strikes Back (at Your Privacy): An Archaeology of Tracking on Government Websites

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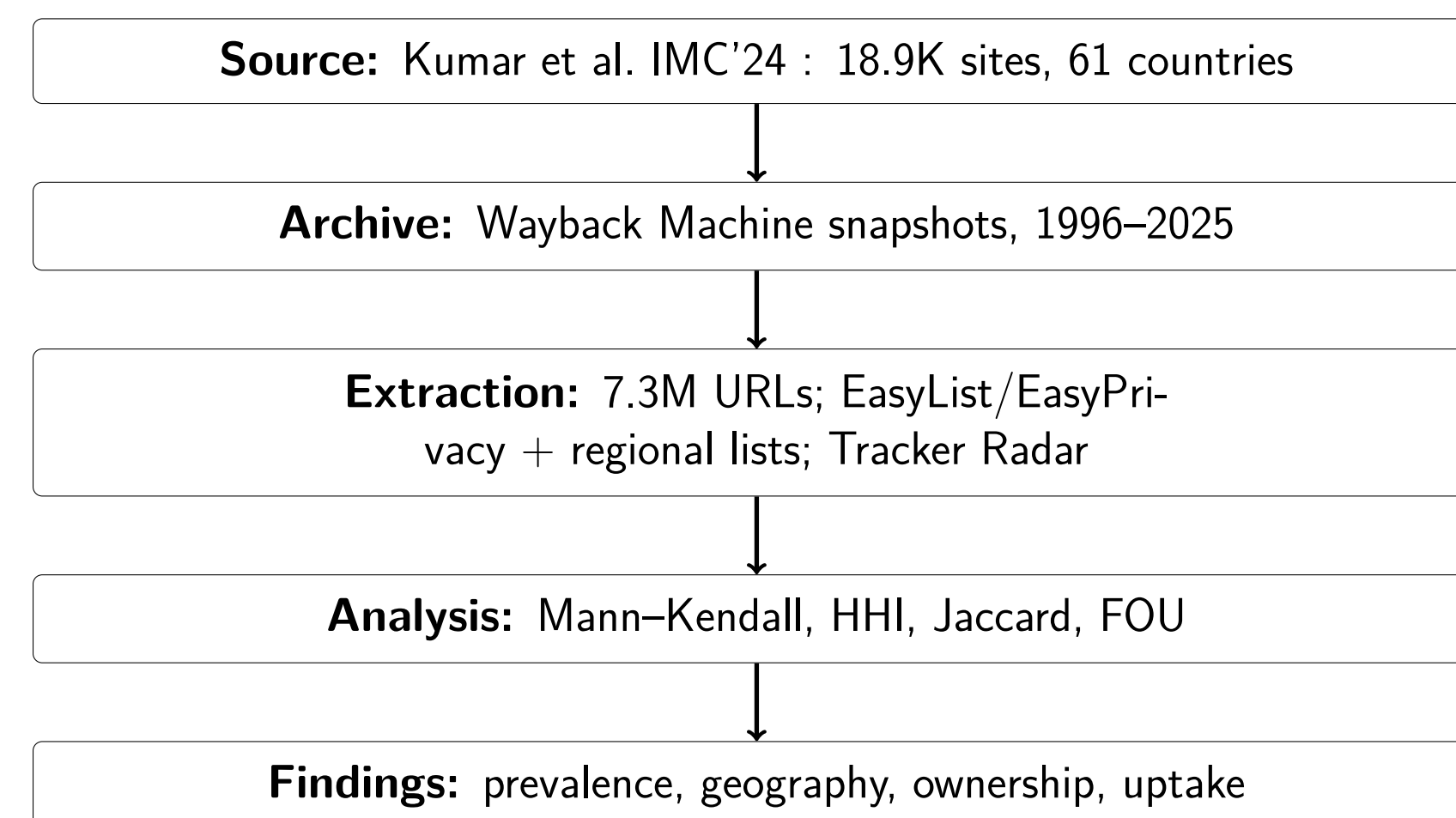
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Motivation

Citizens rely on government websites for essential services like applying for a passport, filing taxes, obtaining healthcare information and expect privacy. Unlike commercial platforms, government portals are often the **only available option**, leaving individuals with no ability to opt out. In practice, many embed third-party trackers that collect browsing behavior and unique identifiers, raising privacy, security, and **data sovereignty** concerns including **cross-border tracker flows**. Prior work focuses on commercial sites or small, single-country government samples and is rarely longitudinal. We present the **first large-scale, global, longitudinal study** of tracking on government websites.

Dataset & Methodology

- 18.9K** federal-level sites across **61 countries**; Internet Archive snapshots **1996–2025**; **7.3M** URLs extracted.
- Tracker labeling**: EasyList + EasyPrivacy + **21 regional filter lists**; ownership via DuckDuckGo Tracker Radar (**365 of 1098** tracker domains mapped to **265 orgs**).
- Statistics**: Mann–Kendall trend tests, Jaccard similarity, HHI, First-Observed Uptake (FOU).



Key Takeaway

Tracking on government websites has shifted from rare to routine: by 2025, trackers appear on $\approx 50\%$ of studied sites. Growth is overwhelmingly driven by **third-party** services (48% of sites vs. only 4.4% first-party), dominated by a handful of **US-based organizations** led by Google and Meta, exposing citizens worldwide to unequal privacy risks on essential public services.

Key Findings

- 50%** of government websites embed a tracker by 2025, up from near zero in 1996; steep post-2016 surge.
- Third-party dominates**: 48% of sites carry third-party trackers vs. only **4.4%** first-party.
- Highly uneven adoption**: Singapore (**95.5%**), Australia (**89%**), Bangladesh (**82.5%**) lead; China (**6.1%**), Germany (**9.2%**), France (**11.6%**) persistently low.
- Concentrated ownership**: Google embedded **68K times** (Meta: 6,635); *google-analytics.com* and *googletagmanager.com* present in **all 61 countries**; mean $HHI_{org} = 0.601$.
- Tracker gaps**: **39 gaps** across **26 countries**; 37 cases suggest deliberate temporary suspensions.
- High churn**: only **30%** of trackers persist year-over-year; avg. **77 new uptakes/year** globally.

Tracker Presence by Country (2025)

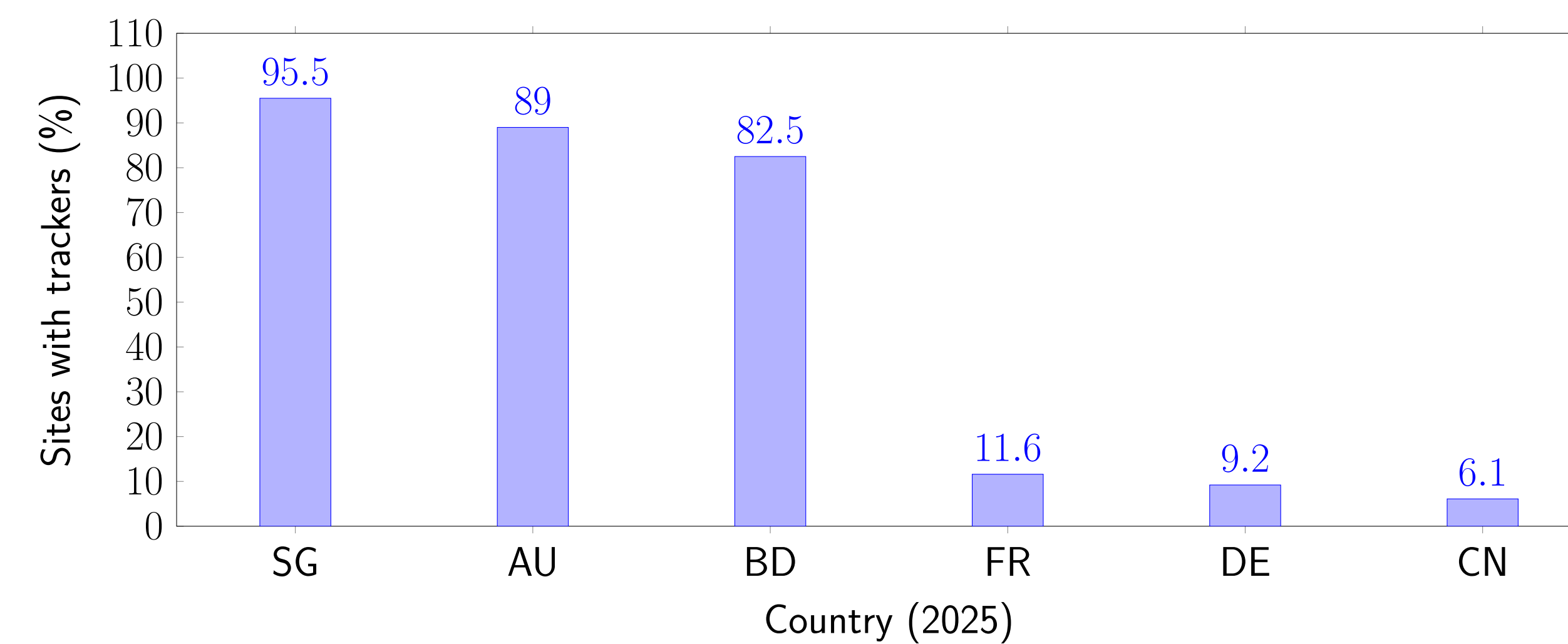


Figure 1. Top 3 vs. bottom 3 countries by tracker presence in 2025.

In Bangladesh, all tracked sites share a **single Google Tag Manager Measurement ID**, indicating a centrally coordinated rollout merging analytics across distinct government services. **26 countries** reached all-time peak tracker presence in 2025, and **14 more** in 2024.

Longitudinal Trends, 1996–2025

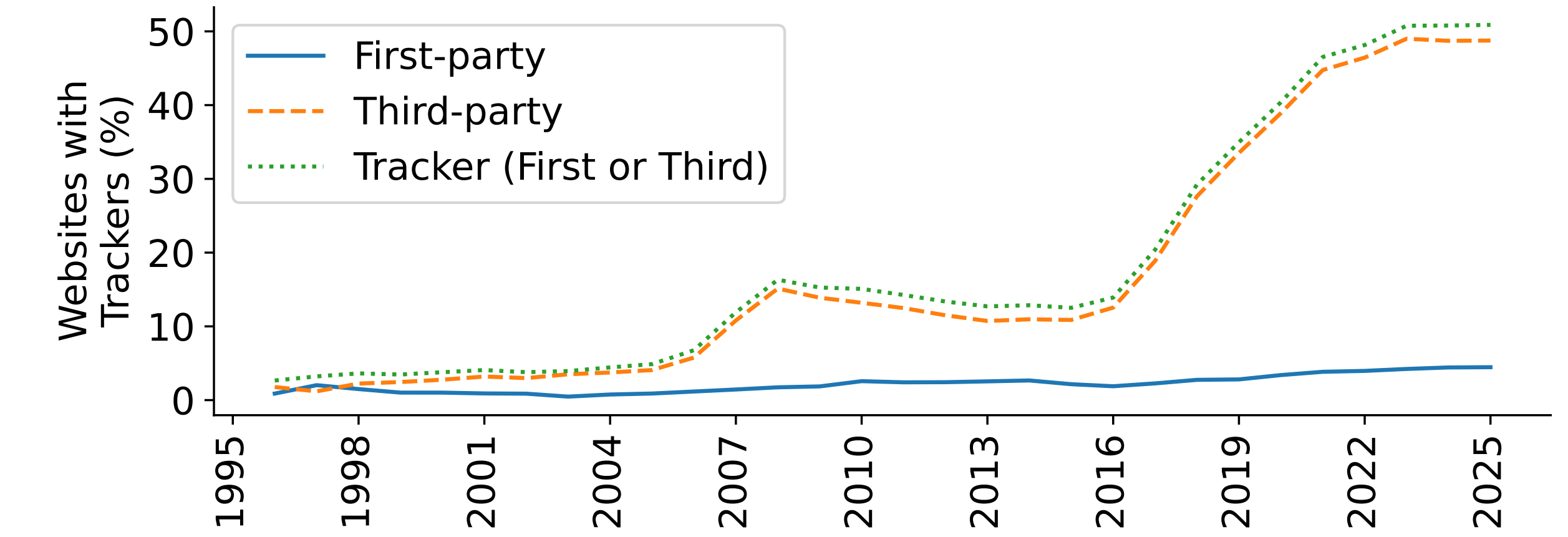


Figure 2. Global tracker presence on government websites, 1996–2025.

